IR: MCP4725 0x60 -> LM358 -> 2N4401 -> LED

RED: MCP4725 0x61 -> LM358 -> 2N4401 -> LED

J1

MCP4725 IR 0x60

J2

PI I2C INPUT / IR OUT

DAC_IR

GND

SCL

SDA

3V3

GND

Module ADDR: GND -> 0x60

Both GND pins share 0 V. OUT is a separate signal.

J2: OUT/GND = 2.54 mm; SCL/SDA/3V3/GND = 2.00 mm

J3

MCP4725 RED 0x61

J4

RED DAC OUT / GND

DAC_RED

GND

SCL

SDA

3V3

GND

Module ADDR: VCC -> 0x61

Both GND pins share 0 V. OUT is a separate signal.

5V

J5

IR LED A / K

LED_K_IR

R1

10k 1%

CMD_IR

U1A

LM358P / DIP8 socket

R3

1k

AMP_IR

BASE_IR

Q1

2N4401 / E-B-C

R2

10k 1%

SENSE_IR

R4

100R 1% 0.25W

GND

I_LED approximately V_DAC / (2 x 100 ohm)

5V

J6

RED LED A / K

LED_K_RED

R6

10k 1%

CMD_RED

U1B

LM358P / DIP8 socket

R8

1k

AMP_RED

BASE_RED

Q2

2N4401 / E-B-C

R7

10k 1%

SENSE_RED

R9

100R 1% 0.25W

GND

I_LED approximately V_DAC / (2 x 100 ohm)

C2

DNP loop comp

AMP_IR

SENSE_IR

C2: DNP – DO NOT FIT until loop measured

C4

DNP loop comp

AMP_RED

SENSE_RED

C4: DNP – DO NOT FIT until loop measured

SHARED SUPPLY: U1A + U1B + U1C = ONE LM358 DIP-8

5V

J7

5V INPUT / GND

GND

C7

10uF 16V

C5

100nF X7R 50V

U1C

LM358P / DIP8 socket

C5: ceramic, non-polar. C7: electrolytic, + to 5V.

H1

M3 / 3.2mm

H2

M3 / 3.2mm

H3

M3 / 3.2mm

H4

M3 / 3.2mm

MCP4725 VCC = 3.3 V ONLY; J2 is the only Pi cable input.

ADDR is configured on each module, not by this carrier PCB.

J2 powers and controls both DAC modules over one shared Pi I2C bus.

OUT pins are DAC outputs: never connect them together.

Matching labels connect SCL, SDA, 3V3 and GND between modules.

For RED: remove the old ADDR-GND bridge before selecting VCC.

Nguyen Nhat Huy – Pham Thanh Vy

Sheet: /

File: led_ir_driver.kicad_sch

Title: DATN: PPG–Simulator – LED / IR driver

Size: A3 | Date: | **Rev: 1.6**

KiCad E.D.A. 10.0.6 | Id: 1/1